

Sheng Kuang

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EDUCATION

Shihezi University

Sept. 2024 – Expected Jun. 2028

B.S. in Biological Science; GPA: 3.51/5.00, Rank: 27/121

Shihezi, Xinjiang, China

Relevant Coursework: Ecology (91.2), Life Science Instruments and Technology (89.7), Plant Morphology and Anatomy (87.6), Biochemistry Experiment (96.3), Molecular Biology (85.1)

Honors: Third-class Scholarship (2025, 2026); Xinjiang TaiKun Group Enterprise Scholarship (2025)

RESEARCH EXPERIENCE

Multi-omics Analysis of Transposable Elements and Their Roles in Salt and Drought Stress Responses in *Glycyrrhiza*

Jan. 2026 – Present

Team Leader

Shihezi University, China

- Led a multi-omics investigation of transposable element (TE) dynamics in three *Glycyrrhiza* species under salt and drought stress, integrating **458,673 TE annotations**, **83 RNA-seq libraries**, and **189 whole-genome resequencing datasets**.
- Developed a custom **Bash pipeline** integrating six structural signals for non-reference TE insertion detection and cross-validated candidate insertions against RetroSeq, identifying an average of $\sim 10,214$ insertions per sample across **176 genomes**.
- Integrated RNA-seq, WGCNA, and machine learning to investigate TE-associated stress responses; developed an interpretable **XGBoost-SHAP** framework with 5-fold cross-validation (**AUC = 0.942**) to prioritize stress-associated TE candidates.
- Awarded Third Prize, National Undergraduate Life Science Competition (Xinjiang Division), 2025.

Genome-Wide Identification of the IFS Gene Family and ARF-Mediated Regulation

Mar. 2025 – Aug. 2025

Undergraduate Researcher, Prof. Shandang Shi & Prof. Shuangquan Xie

Shihezi University, China

- Conducted genome-wide identification and comparative characterization of the isoflavone synthase (IFS) gene family across three *Glycyrrhiza* species (*G. uralensis*, *G. glabra*, and *G. inflata*), identifying **28 IFS genes** through BLAST-based homology searches and Pfam/NCBI-CDD domain verification.
- Reconstructed phylogenetic relationships using MEGA X with 1,000 bootstrap replicates and integrated chromosomal localization, gene structure, conserved motif, and synteny analyses to characterize the evolutionary history of the IFS family.
- Analyzed promoter regions using PlantCARE and identified hormone- and stress-responsive cis-regulatory elements associated with auxin, ABA, MeJA, drought, and low-temperature responses, providing candidate regulatory mechanisms for isoflavonoid biosynthesis.
- Integrated computational analyses with experimental validation in collaboration with wet-lab researchers; co-authored a peer-reviewed publication in *International Journal of Molecular Sciences* (2025, 26, 10435).

Cold Stress Tolerance in Cotton: From Gene Function to Commercial Application

Mar. 2025 – Mar. 2026

Undergraduate Researcher, Prof. Hongbin Li

Shihezi University, China

- Contributed to functional characterization of *GhZF-HD19*, a zinc finger-homeodomain transcription factor, in *Gossypium hirsutum* cold tolerance through phenotypic evaluation of overexpression and VIGS-silenced plants under chilling stress.
- Performed physiological assays measuring MDA, SOD, POD, proline, chlorophyll, soluble sugars, and soluble proteins to evaluate membrane stability, antioxidant capacity, and physiological responses to cold stress.
- Translated experimental findings into a competitive innovation project based on patented cold-tolerance technology (Patent No. 2025-6515), contributing to a **First Prize** at the National Cotton and Hemp Innovation and Entrepreneurship Competition.

Computational Protein Design for Synthetic Biology

Jan. 2026 – Aug. 2026

Team Leader / Team Member, Protein Design Track

Shenzhen, China

- Collaborated on computational protein engineering using **AlphaFold** structural prediction and **Rosetta** energy minimization to design and evaluate protein variants, followed by experimental validation and site-directed mutagenesis.
- Led a project team to optimize engineered protein variants through molecular cloning, heterologous expression in *E. coli*, and biochemical assays; coordinated experimental design, data analysis, project scheduling, and scientific presentation.
- Received **Gold Award** as a team member and **Silver Award** as team leader at the 5th SynBio Challenges (Synthetic Biology Innovation Competition), Shenzhen, 2026.

Student Achievement Visualization Teaching Evaluation Platform

Feb. 2026 – Aug. 2026

Co-Principal Investigator (Second Leader), supervised by Prof. Xia Bo & Prof. Xu Liming

Shihezi University, China

- Co-led the development of a full-stack educational assessment platform implementing automated four-level weighted aggregation

of course objectives and graduation requirements.

- Designed PostgreSQL database schemas and RESTful APIs using Node.js/Express, and implemented role-based access control with tag-level data isolation.
- Developed ECharts-based competency visualization and batch PDF reporting, and containerized the platform with Docker Compose for cloud deployment and pilot testing.
- Awarded **National Third Prize** at the 19th China Undergraduate Computer Design Competition (Software Application and Development), 2026.

PUBLICATIONS

Xu, Q.; Hu, X.; Cui, S.; Gao, J.; Zeng, L.; Li, Z.; **Kuang, S.**; Chen, X.; Xie, Q.; Li, Z.; Li, H.; Wang, F.; Shi, S.; Xie, S. Genome-Wide Identification and Characterization of Isoflavone Synthase (IFS) Gene Family, and Analysis of GgARF4-GgIFS9 Regulatory Module in *Glycyrrhiza glabra*. *International Journal of Molecular Sciences* **2025**, 26, 10435. DOI: [10.3390/ijms262110435](https://doi.org/10.3390/ijms262110435).

HONORS & AWARDS

2026 Gold Award, 5th SynBio Challenges — Protein Design Track, Shenzhen, China; **Silver Award** as Team Leader

2026 First Prize, National Cotton and Hemp Innovation and Entrepreneurship Competition

2026 National Third Prize, 19th China Undergraduate Computer Design Competition — Software Application and Development, as Second Team Leader

2026 Third Prize, National Undergraduate Life Science Competition (Xinjiang Division)

2026 Third Prize, National English Competition for College Students (NECCS), Xinjiang Division

2026 First Prize, Campus Eco-Civilization Creative Works Competition

2025 Outstanding Student Cadre, Shihezi University

TECHNICAL SKILLS

Bioinformatics RNA-seq analysis (HISAT2, featureCounts, DESeq2); TE annotation ; phylogenetic analysis ; synteny and genomic analysis ; WGCNA; BEDTools

Programming R, Python, Bash/Shell scripting, C++/C, Java, HTML/CSS, JavaScript,

Machine Learning XGBoost, SHAP, cross-validation, and model evaluation

Molecular Biology DNA/RNA extraction, PCR, gel electrophoresis, molecular cloning, heterologous expression in *E. coli*, site-directed mutagenesis, biochemical assays

Web Development Vue3, Node.js/Express, PostgreSQL, Docker, RESTful API design